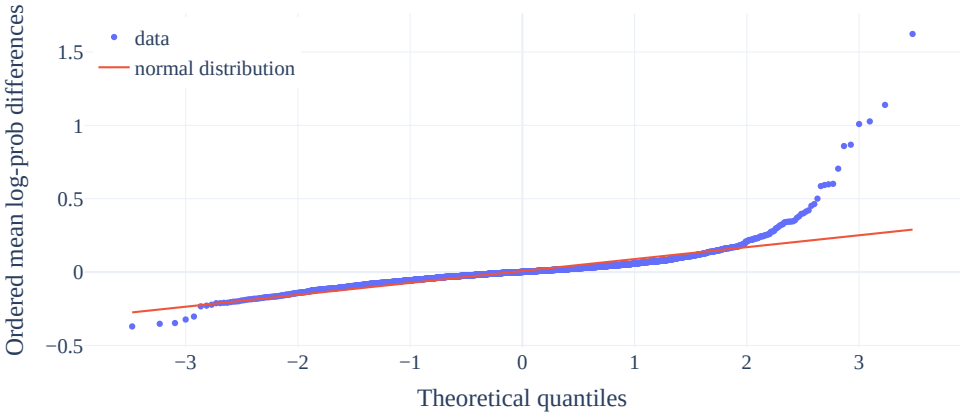
Table 11: Tokens with the greatest absolute change in log probability under ActAdd(`weddings`). (See Figure 8 for the distribution these are drawn from.) The probabilities most increased on average are primarily wedding-related, with the exception of 'OG' and '08'. (We conjecture that their representations are in 'superposition' with wedding-related tokens Elhage et al. 2022). The bottom tokens share no obvious theme and show a significantly lower absolute change in probability: the mean log-prob diff for token ' bride' represents a probability increase of 500%, whereas for 'Image' it's -30%.

token	mean_logprob_diff	mean_logprob_normal
marry	0.593	-3.509
dress	0.598	-5.692
dating	0.601	-6.891
08	0.705	-10.749
married	0.859	-4.613
OG	0.868	-11.287
weddings	1.009	-6.698
wedding	1.027	-4.593
br	1.139	-6.438
bride	1.623	-6.652
Image	-0.370	-1.836
.)	-0.352	-2.378
BP	-0.347	-7.897
U+25CF	-0.323	-0.201
Apple	-0.303	-5.058
On	-0.233	-5.404
journalists	-0.229	-4.484
defense	-0.222	-4.864
Russian	-0.212	-5.112
It	-0.212	-6.431

Figure 8: Distribution shift (in mean log-probability changes) under ActAdd, relative to the unmodified model, and compared to a normal distribution's quantiles (red). The resulting distribution is approximately normal for most tokens. The positive tail is significantly heavier than the negative tail: one set of tokens are reliably increased in probability, one reliably decreased. See Appendix Table 11 for the corresponding tokens.



C.1 Details of perplexity experiments

For each sentence in each document, we calculate the log-probabilities $\mathcal{L}(t_k)$ for each token $t_k \in s_j$ under the unmodified M_{baseline} and modified M_{ActAdd} models.

We compute the mean token log-probability $\bar{\mathcal{L}}(d_i, M)$ for each document and model. We then group documents by their wedding-word frequency f_w (e.g. 'those with 0.5% to 1% of their tokens wedding-related'; 'those with 1 to 1.5% of their tokens wedding-related'), producing bins of documents b_m. We calculate the mean difference in token log-probabilities

$\bar{X}(b_m) = \text{mean}_{d_i \in b_m} \left(\bar{\mathcal{L}}(d_i, M_{\text{ActAdd}}) - \bar{\mathcal{L}}(d_i, M_{\text{baseline}})\right)$ for each bin. (We use only bins with a number of documents $|b_m| > 1000$, to reduce sampling noise.) Finally, the change in perplexity under ActAdd for each wedding-word-frequency bin is $\text{PerplexityRatio}(b_m) = -\exp(\bar{X}(b_m))$.

D Advantages of ActAdd over fine-tuning

Following Sloman 2002, we distinguish 'ballistic' steering (which steers the model once, e.g. at train time) from 'online' steering (which can steer the model repeatedly, e.g. at inference time). Fine-tuning is ballistic, while ActAdd is online in this sense - which enables iteration and otherwise infeasible chains and mixes of steering decisions.

Activation additions may preserve model interpretability, even while changing the model's alignment. When finetuning a model, a single gradient update can change every parameter in it, thereby undoing your prior interpretability work, which depends on tracking individual neurons and circuits of neurons. By contrast, activation additions leave weights unchanged. If we understand what algorithms the weights implement, and something about the effects of our activation additions, we will preserve our understanding of the steered model. Finally, we hypothesize that activation addition may allow control over properties inaccessible to the fine-tuning process.

The intuition is that since the *currently-active* goal is contextual, it depends more on short-lived activations than the weights (which instead represent some analogue of skills and other stable patterns and *mixtures* of possible goals).

Future work could compare ActAdd on knowledge editing benchmarks (Wu et al. 2023; Zhang et al. 2024. (This comparison could be unfair to methods like ROME Meng et al. 2023), since our method is not editing weights, but it would give standardized evidence about steering.)

Prompt	Baseline	Random steering
I think you're	...right. I'm just not sure how to get there. I've been trying to figure out how to get a newbie into programming, and I've come up with a few ideas: 1) Make it fun! If you can't do something that's fun, then you	...right. I'm just not sure how to make it work. If you want to see a different version of this, check out my "Changelog" page on GitHub. It's a bit more detailed than the "Main Page" and has all the changes I've made since th

Table 12: Example of a random-vector ActAdd. We see little qualitative effect, over many runs. See Figure 10 for a systematic look at the relative size of output token distribution shift.

E REPLICABILITY

We now check that ActAdd steering generalizes to models besides GPT-2.

E.1 GPT-J-6B

Figures 11, 12, and 13 show the results from repeating the main experiments on GPT-J-6B Wang & Komatsuzaki 2021. We see the same dynamics from the wedding vector running example: a targeted effect on only wedding-related tokens (using both KL-div and token probability); and similar effects when injected at different layers of GPT-J and with different magnitudes c applied.

E.2 LLAMA-1-13B

Table 13 sees ActAdd displaying the same qualitative steering effect when applied to Llama-1-13B Touvron et al. 2023 (though with a notable failure to replicate on Example 6, Paris $\rightarrow$ Rome, the anger vector, and the harm vector).

E.3 OPT-6.7B

We use the OPT model Zhang et al. 2022b in our toxicity (Table 3) and sentiment (Table 4) experiments. ActAdd-OPT using the love$-$hate vector produces a statistically significant 17% drop in toxicity over an unsteered OPT, at a small (partially unavoidable owing to the nature of the detoxification task) cost to fluency and relevance. ActAdd-OPT using the love$-$hate vector produces a 21% absolute increase in positive classification over an unsteered OPT, at a larger (partially unavoidable owing to the nature of the sentiment shift task) cost to fluency and relevance.

E.4 LLAMA-3-8B

We also use Llama-3-8B Meta 2024 in our toxicity (Table 3) and sentiment (Table 4) experiments. ActAdd-LLaMA-3 using the love$-$hate vector produces a statistically significant 5% drop in toxicity over an unsteered Llama-3-8B, at a very small (partially unavoidable owing to the nature of the detoxification task) cost to fluency and relevance. ActAdd-LLaMA-3 using the love$-$hate vector produces a 25% absolute increase in negative-to-positive classification over an unsteered Llama-3-8B, at a larger (partially unavoidable owing to the nature of the sentiment shift task) cost to fluency and relevance.

F INVESTIGATING THE NORM OF STEERING VECTORS

Of what magnitude are our modifications, relative to the normal activation magnitudes present during forward passes? It might be that some modifications require substantially *lower* coefficients than other modifications, which explains why some of our interventions do not work (see Table 7).

Consider the steering vector given by

$$\{c = +1,\, p_+ = \text{anger},\, p_- = \text{calm},\, l = 20,\, p^* = \text{I think you're}\}$$